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Silverio Martínez Andrés

Algebra Lineal

Grupo 17

Semestre 2022-2

Tarea 11

Sea P_1 el espacio vectorial real de los polinomios de grado menor o igual a uno con coeficientes reales. La matriz asociada a la transformación lineal $T: P_1 \rightarrow R^3$ referida a las bases

$$A = \{2x+1, -x+1\}$$

$$B = \{(0,0,1), (0,1,1), (1,1,1)\}$$

es

$$M_B^A(T) = \begin{pmatrix} -5 & 1 \\ 4 & 1 \\ -1 & -1 \end{pmatrix}$$

Obtener:

a) La regla de correspondencia de T.

b) El núcleo y el recorrido de T.

Solucion

a)

$$M_B^A(T)\,[\overline{V}]_A = [T(\overline{V})]_B$$

$\overline{V}$: $(ax+b)$; $T(\overline{V})$: (x,y,z)

$$M_B^A(T)\,[(ax+b)]_A = [T(x,y,z)]$$

$$[(ax+b)]_A$$

$$(ax+b) = \alpha_1(2x+1) + \alpha_2(-x+1)$$

$$(ax+b) = (2x\alpha_1 - x\alpha_2,\ \alpha_1+\alpha_2)$$

$$\left.\begin{aligned} 2x\alpha_1 - x\alpha_2 &= ax \\ \alpha_1+\alpha_2 &= b \end{aligned}\right\}\text{sist. d. ecu.}$$

$$\left(\begin{array}{cc|c} 2 & -1 & a \\ 1 & 1 & b \end{array}\right) R_1 \Leftrightarrow R_2 \quad \left(\begin{array}{cc|c} 1 & 1 & b \\ 2 & -1 & a \end{array}\right) R_2 + R_1 = R_1$$

$$\left(\begin{array}{cc|c} 3 & 0 & a+b \\ 2 & -1 & a \end{array}\right) \begin{matrix} R_1(\frac{1}{3}) \\ R_2(-1) \end{matrix} \quad \left(\begin{array}{cc|c} 1 & 0 & \frac{a+b}{3} \\ -2 & 1 & -a \end{array}\right) 2R_1 + R_2 = R_2$$

$$\left(\begin{array}{cc|c} 1 & 0 & \frac{a+b}{3} \\ 0 & 1 & \frac{-a+2b}{3} \end{array}\right)$$

$$[(ax+b)]_A = \begin{pmatrix} \frac{a+b}{3} \\ \frac{-a+2b}{3} \end{pmatrix}$$

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$$\begin{pmatrix} -5 & 1 \\ 4 & 1 \\ -1 & -1 \end{pmatrix}\begin{pmatrix} \frac{a+b}{3} \\ \frac{-a+2b}{3} \end{pmatrix} = [T(x,y,z)]_B$$

$$\begin{pmatrix} -2a-b \\ a+2b \\ -b \end{pmatrix} = [T(x,y,z)]_B$$

$$B = \{(0,0,1),(0,1,1),(1,1,1)\}$$

$$T(x,y,z) = \alpha_1\bar{w}_1 + \alpha_2\bar{w}_2 + \alpha_3\bar{w}_3$$

$$T(x,y,z) = (-2a-b)(0,0,1) + (a+2b)(0,1,1) + (-b)(1,1,1)$$

$$T(x,y,z) = (0,0,-2a-b) + (0,a+2b,a+2b) + (-b,-b,-b)$$

$$T(x,y,z) = (-b,\ a+2b-b,\ -2a-\cancel{b}+a+\cancel{2b}-\cancel{b})$$

$$T(x,y,z) = (-b,\ a+b,\ -a)$$

b)

$$N(T) = \{\bar{v}\in V \mid T(\bar{v}) = 0_W;\ 0_W \in W\}$$

$$N(T) = \{\underbrace{(x,y,z)\in R^3}_{2°} \mid \underbrace{T(x,y,z) = (0,0,0)}_{1°};\ (0,0,0)\in R^3\}$$

$T(x, y, z) = (0, 0, 0)$

$(-b, a+b, -a) = (0, 0, 0)$

$$\left.\begin{aligned} -b &= 0 \\ a+b &= 0 \\ -a &= 0 \end{aligned}\right\} \text{sis ec}$$

$a + b = 0$ $\qquad$ $-(-a) = 0$

$\underline{a = -b}_{\,/\!/}$ $\qquad$ $\underline{a = 0}_{\,/\!/}$

$a + b = 0$ $\qquad$ $-(-b) = 0$

$b = -a$ $\qquad$ $\underline{b = 0}_{\,/\!/}$

$N(T) = \{(x, y, z) \in \mathbb{R}^3\}$

$N(T) = \{(0, 0+0, 0) \in \mathbb{R}^3\}$

X

2°

$IM(T) = \{\bar{\omega} \in \omega \mid T(\bar{v}) = \bar{\omega};\ \bar{v} \in V\}$

$IM(T) = \{(m, n, p) \in \mathbb{R}^3 \mid T(x, y, z) = (m, n, p);\ x, y, z \in \mathbb{R}^3\}$

$T(x, y, z) = (m, n, p)$

$(-b, a+b, -a) = (m, n, p)$

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$$\left.\begin{array}{l} -b = m \\ a + b = n \\ -a = p \end{array}\right\} \begin{array}{l} s \\ d \\ c \end{array}$$

$-b = m$

$\underline{b = -m}_{\#}$

$-a = p$

$\underline{a = -p}_{\#}$

$a + b = n$

$\underline{-p - m = n}_{\#}$

$$IM(T) = \{(m, n, p) \in R^3\}$$

$$IM(T) = \{(-m, -m-p, -p) \mid m, p \in R^3\}$$

$\frac{1}{2}$